Mth202 Short Notes Mid term

Lec 1 to22

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آپ کو کبھی بھی وہی جواب کاپی پیسٹ نہیں کرنا چاہیے جو ہم فراہم کر رہے ہیں۔ براہ کرم حل فائل میں کچھ تبدیلیاں کریں۔ اگر آپ ہمارے کام کی تعریف کرتے ہیں تو براہ کرم ہماری ویب سائٹ vuonlinehelp.blogspot.com کو اپنے دوستوں کے ساتھ شیئر کریں

Lecture No.1 Logic

What is Discrete Mathematics?

Discrete Mathematics concerns processes that consist of a sequence of individual steps.

Define LOGIC:

Logic is the study of the principles and methods that distinguish between a valid and an invalid argument.

What is SIMPLE STATEMENT?

A statement is a declarative sentence that is either true or false but not both. A statement is also referred to as a proposition.

Define Logical Connective?

There are five logical connectives

A Logical Connective is a symbol which is used to connect two or more propositional or predicate logics in such a manner that resultant logic depends only on the input logics and the meaning of the connective used. Generally there are five connectives which are − OR (∨) AND (∧)

SYMBOLIC REPRESENTATION

CONNECTIVE	MEANINGS	SYMBOLS	CALLED
Negation	not	~	Tilde
Conjunction	And/but	∧	Hat
Disjunction	or	∨	Vel
Conditional	If...then...	→	Arrow
Biconditional	if and only if	↔	Double arrow

EXAMPLE

SENTENCE SYMBOLIC FORM

1. It is **not** hot. ~ p

2. It is hot **and** sunny. P ∧q

3. It is hot **or** sunny. P ∨ q

4. It is **not** hot **but** sunny. ~ P ∧q

5. It is **neither** hot **nor** sunny. ~ P ∧ ~ q

Define Truth Table?

A truth table specifies the truth value of a compound proposition for all possible truth values of its constituent propositions.

Define NEGATION (~)?

If p is a statement variable, then negation of p, "not p", is denoted as "~p"

TRUTH TABLE FOR ~ p

p	~ p
T	F
F	T

Define CONJUNCTION (∧)?

If p and q are statements, then the conjunction of p and q is "p and q", denoted as "p ∧ q".

TRUTH TABLE FOR p ∧ q.

p	q	p ∧ q
T	T	T
T	F	F
F	T	F
F	F	F

What is DISJUNCTION (V) or INCLUSIVE OR?

If p & q are statements, then the disjunction of p and q is "p or q", denoted as "p V q".

TRUTH TABLE FOR p V q

p	q	p v q
T	T	T
T	F	T
F	T	T
F	F	F

Lecture No.3 Laws of Logic

WHAT IS CONDITIONAL STATEMENTS?

The original statement is then saying:

If p is true, then q is true, or, more simply, if p, then q. We can also phrase this as p implies q. It is denoted by p → q.

p	q	p → q
T	T	T
T	F	F
F	T	T
F	F	T

FOR LOGICAL CONNECTIVES

•~ (negation)

•∧ (conjunction), ∨ (disjunction)

•→ (conditional)

INVERSE OF A CONDITIONAL STATEMENT:

The inverse of the conditional statement p → q is ~p → ~q A conditional and its inverse are not equivalent as could be seen from the truth table.

P	q	P → q	~p	~q	~p → ~q
T	T	T	F	F	T
T	F	F	F	T	T
F	T	T	T	F	F
F	F	T	T	T	T

CONVERSE OF A CONDITIONAL STATEMENT:

The converse of the conditional statement p → q is q →p. A conditional and its converse are not equivalent. i.e., → is not a commutative operator.

P	Q	P → q	Q → p
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

Lecture No. 4 Biconditional

BICONDITIONAL

If p and q are statement variables, the biconditional of p and q is “p if and only if q”. It is denoted $p \leftrightarrow q$. “if and only if” is abbreviated as iff.

The double headed arrow “$\leftrightarrow$" is the biconditional operator.

P	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Lecture No. 5 Argument

ARGUMENT:

An argument is a list of statements called premises (or assumptions or hypotheses) followed by a statement called the conclusion.

P1 Premise

P2 Premise

P3 Premise

. . . .

p_n Premise

____

∴C Conclusion

VALID AND INVALID ARGUMENT:

An argument is valid if the conclusion is true when all the premises are true. An argument is invalid if the conclusion is false when all the premises are true.

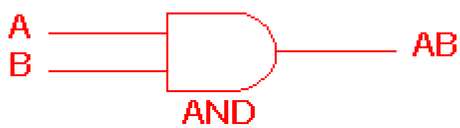
Critical Rows:

The critical rows are those rows where the premises have truth value T

Lecture No.6 Applications of Logic

AND gate

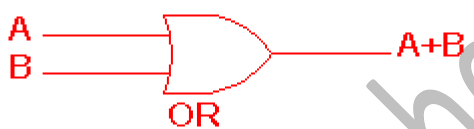
The AND gate is an electronic circuit that gives a **high** output (1) only if all its inputs are high. A dot (.) is used to show the AND operation i.e. A.B. Bear in mind that this dot is sometimes omitted i.e. AB



A	B	AB
0	0	0
0	1	0
1	0	0
1	1	1

OR gate

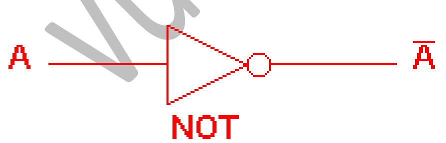
The OR gate is an electronic circuit that gives a high output (1) if **one or more** of its inputs are high. A plus (+) is used to show the OR operation.



A	B	AB
0	0	1
0	1	1
1	0	1
1	1	0

NOT gate

NOT-gate A NOT-gate (or inverter) is a circuit with one input and one output signal. If the input signal is 1, the output signal is 0. Conversely, if the input signal is 0, then the output signal is 1.



NOT gate	
A	$\overline{A}$
0	1
1	0

Lecture No.7 Set theory

TABULAR FORM

We list all the elements of a set, separated by commas and enclosed within braces or curly brackets {}.

NOTE: The symbol "…" is called an ellipsis. It is a short for "and so forth."

ESCRIPTIVE FORM:

We state the elements of a set in words.

E,g

A = set of first five Natural Numbers. (Descriptive Form)

SET BUILDER FORM:

We write the common characteristics in symbolic form, shared by all the elements of the set.

E,g
$C = \{ x \in O \mid 0 < x \}$ (Set Builder Form)

SUBSET: $\subseteq$

If A & B are two sets, then A is called a subset of B. It is written as $A \subseteq B$. The set A is subset of B if and only if any element of A is also an element of B. Symbolically:

$$A \subseteq B \Leftrightarrow \text{if } x \in A, \text{ then } x \in B$$

REMARK:

1. When $A \subseteq B$, then B is called a superset of A.
2. When A is not subset of B, then there exist at least one $x \in A$ such that $x \notin B$.
3. Every set is a subset of itself.

PROPER SUBSET: $\subset$

Let A and B be sets. A is a proper subset of B, if and only if, every element of A is in B but there is at least one element of B that is not in A, and is denoted as $A \subset B$.

NULL SET:

A set which contains no element is called a null set, or an empty set or a void set. It is denoted by the Greek letter Ø (phi) or { }.

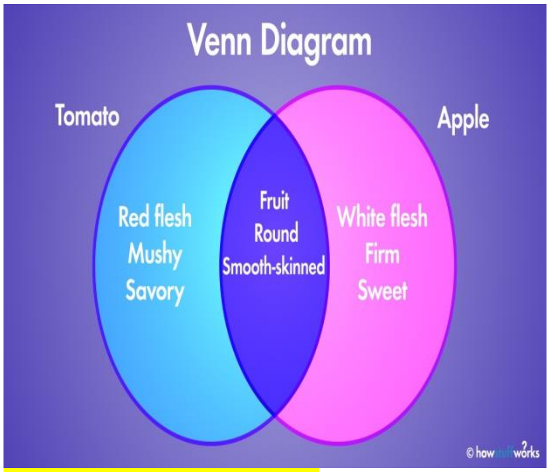
REMARK Ø is regarded as a subset of every set.

UNIVERSAL SET:

The set of all elements under consideration is called the Universal Set. The Universal Set is usually denoted by U.

VENN DIAGRAM:

A Venn diagram is a graphical representation of sets by regions in the plane. The Universal Set is represented by the interior of a rectangle, and the other sets are represented by disks lying within the rectangle.



FINITE AND INFINITE SETS:

A set S is said to be finite if it contains exactly m distinct elements where m denotes some non-negative integer.

In such case we write |S| = m or n(S) = m

A set is said to be infinite if it is not finite.

E,g

1. A = {month in the year} FINITE

2. B = {even integers} INFINITE

MEMBERSHIP TABLE:

If an element is not a member of A, it will be the member of Ac. So we have 0 for A and 1 for Ac.

A	Ac
1	0
0	1

Lecher number # 8 Venn diagram

Unions

An element is in the union of two sets if it is in the first set, the second set, or both. The symbol we use for the union is ∪.

$A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}$

INTERSECTION:

Let A and B subsets of a universal set U. The intersection of sets A and B is the set of all elements in U that belong to both A and B and is denoted $A \cap B$.

Symbolically: $A \cap B = \{x \in U \mid x \in A \text{ and } x \in B\}$

COMPLEMENT:

Let A be a subset of universal set U. The complement of A is the set of all element in U that do not belong to A, and is denoted AN, A or Ac

Symbolically: $Ac = \{x \in U \mid x \notin A\}$

Lecture No.9 Set identities

Only practice's qustion

Lecture No.10 Applications of Venn diagram

PARTITION OF A SET

A set may be divided up into its disjoint subsets. Such division is called a partition

A partition of a set A is a collection of non- empty subsets {A1, A2...... An} of A, such that

1. $A = A_1 \cup, A_2 \cup \dots \cup A_n$
2. A1, A2, ..., An are mutually disjoint (or pair wise disjoint),

POWER SET:

The power set of a set A is the set of all subsets of A, denoted P (A).

E,g

Let A = {1, 2},

Then

P (A) = {Ø, {1}, {2}, {1, 2}}

Lecture No.11 Relations

ORDERED PAIR:

The ordered pairs (a, b) and (c, d) are equal if, and only if, a= c and b = d. Note that (a, b) and (b, a) are not equal unless a = b.

EXERCISE:

Find x and y given (2x, x + y) = (6, 2)

Find value of x then putting in 2nd eq

An ordered n-tuple

A sequential arrangement of **n** of objects **order** matters, the same object may occur at different positions. A point in 3-dimensional space is represented as an **ordered** 3-**tuple**/triple x, y, z. In a relational database, data is stored in tables.

CARTESIAN PRODUCT OF TWO SETS

Let A and B be sets. The Cartesian product of A and B, denoted by A × B (read as "A cross B") is the set of all ordered pairs (a, b), where a is in A and b is in B. Symbolically:

$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$

EXAMPLE:

Let A = {1, 2}, B = {a, b, c}

Then

A × B = {(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)} B × A = {(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)}

A × A = {(1, 1), (1,2), (2, 1), (2, 2)}

B × B = {(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b),(c, c)}

BINARY RELATION:

Let A and B be sets. The binary relation R from A to B is a subset of A × B. When (a, b) ∈R, we say 'a' is related to 'b' by R, written aRb.

Otherwise, if (a, b) ∉R, we write a R b.

EXAMPLE:

Let A = {1, 2},   B = {1, 2, 3}

Then A × B = {(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)}

Let R1= {(1, 1), (1, 3), (2, 2)}

R2= {(1, 2), (2, 1), (2, 2), (2, 3)}

R3= {(1, 1)}

R4= A × B

R5= ∅

All being subsets of A × B are relations from A to B.

DOMAIN OF A RELATION:

The domain of a relation R from A to B is the set of all first elements of the ordered pairs which belong to R denoted by Dom(R). Symbolically,

Dom (R) = {a ∈A | (a, b) ∈R}

RANGE OF A RELATION:

The range of a relation R from A to B is the set of all second elements of the ordered pairs which belong to R denoted Ran(R). Symbolically,

Ran(R) = {b ∈B | (a, b) ∈ R}

RELATION ON A SET:

A relation on the set A is a relation from A to A. In other words, a relation on a set A is a subset of A × A.

For any set A

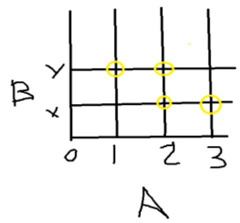
1. A × A is known as the universal relation.

2. ∅ is known as the empty relation.

COORDINATE DIAGRAM (GRAPH) OF A RELATION:

Let A = {1, 2, 3} and B = {x, y}

Let R be a relation from A to B defined as



R = {(1, y), (2, x), (2, y), (3, x)}

MATRIX REPRESENTATION OF A RELATION

Let A = {a1, a2, …, an} and B = {b1, b2, …, bm}. Let R be a relation from A to B. Define the n × m order matrix M by

Lecture No.12 Types of Relations

REFLEXIVE RELATION:

Let R be a relation on a set A. R is reflexive if and only if, for all a ∈ A, (a, a) ∈R or equivalently aRa. That is, each element of A is related to itself.

R1 = {(1, 1), (3, 3), (2,2), (4, 4)}

R1 is reflexive, since (a, a) ∈R1 for all a ∈A.

MATRIX REPRESENTATION

OF A REFLEXIVE RELATION: Let A = {a1 , a2, …, an }. A Relation R on A is reflexive if and only if (ai , aj) ∈R ∀ i=1,2, …,n. Accordingly, R is reflexive if all the elements on the main diagonal of the matrix M representing R are equal to 1.

SYMMETRIC RELATION

Let R be a relation on a set A. R is symmetric if, and only if, for all a, b ∈ A, if (a, b)∈R, then (b, a) ∈R. That is, if aRb then bRa.

R1 = { (1, 3), (2, 4), (3, 1), (4,2)}

Then R1 is symmetric because for every order pair (a, b) in R1also have (b, a) in R1.

For example, we have (1, 3) in R1 then we have (3, 1) in R1. Similarly all other ordered pairs can be checked.

TRANSITIVE RELATION

Let R be a relation on a set A. R is transitive if and only if for all a, b, c ∈A, if (a, b) ∈R and (b, c) ∈R then (a, c) ∈R. That is, if aRb and bRc then aRc.

Let A = {1, 2, 3, and 4} and define relations R1, R2 and R3 on A as follows:

R1 = {(1, 1), (1, 2), (1, 3), (2, 3)}

R2 = {(1, 2), (1, 4), (2, 3), (3, 4)}

R3 = {(2, 1), (2, 4), (2, 3), (3, 4)}

Then R1 is transitive because (1, 1), (1, 2) are in R, then to be transitive relation (1,2) must be there and it belongs to R.

Similarly for other order pairs. R2 is not transitive since (1,2) and (2,3) ∈ R2 but (1,3) ∉ R2.

R3 is transitive.

Lecture No.13 Matrix Representation of Relations

IRREFLEXIVE RELATION:

Let R be a binary relation on a set A. R is irreflexive iff for all a∈A,(a,a) ∉R. That is, R is irreflexive if no element in A is related to itself by R.

R1 = {(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)}

Then R1 is irreflexive since no element of A is related to itself in R1. i.e. (1, 1) ∉ R1, (2,2) ∉ R1 , (3,3) ∉ R1 ,(4,4) ∉ R1

ANTISYMMETRIC RELATION:

Let R be a binary relation on a set A.R is anti-symmetric iff ∀a, b ∈A if (a,b) ∈R and (b,a) ∈R then a = b.

Let A = {1, 2, 3, 4} and define the following relations on A.

R1 = {(1, 2), (2, 2), (2, 3), (3, 4), (4, 1)}

R1 is anti-symmetric but not symmetric because (1, 2) ∈ R1but (2, 1) ∉ R1.

PARTIAL ORDER RELATION:

Let R be a binary relation defined on a set A. R is a partial order relation,if and only if, R is reflexive, antisymmetric, and transitive. The set A together with a partial ordering R is called a partially ordered set or poset.

Lecture No.14 Inverse of Relations

INVERSE OF A RELATION:

Let R be a relation from A to B. The inverse relation R-1 from B to A is defined as: R-1 = {(b,a) ∈B×A | (a,b) ∈R}

COMPOSITE RELATION:

Let A, B, and C be sets, and let R be a relation from A to B and let S be a relation from B to C. That is, R is a subset of A × B and S is a subset of B × C. Then R and S give rise to a relation from A to C indicated by R◦S and defined by: a (R◦S)c if for some b ∈ B we have aRb and bSc.

EXAMPLE:

Define R = {(a,1), (a,4), (b,3),(c,1), (c,4)} as a relation from A to B and S = {(1,x),(2,x), (3,y), (3,z)} be a relation from B to C.

A(1 2) b(3 4) C (5 6)

R=a *b

S=b*c

Lecture No.15 Functions

RELATIONS AND FUNCTIONS:

In mathematics, a function can be defined as a rule that relates every element in one set, called the domain, to exactly one element in another set, called the range. For example, $y = x + 3$ and $y = x^2 – 1$ are functions because every x-value produces a different y-value. A relation is any set of ordered-pair numbers.

Which of the relations define functions from X = {2,4,5} to Y={1,2,4,6}. a.

R1 = {(2,4), (4,1)}

a. R1 is not a function, because 5 ∈X does not appear as the first element in any ordered pair in R1.

FUNCTION:

A function f from a set X to a set Y is a relationship between elements of X and elements of Y such that each element of X is related to a unique element of Y, and is denoted f : X →Y. The set X is called the domain of f and Y is called the co-domain of f.

Graph of $y = x^2$

BINARY OPERATION AS FUNCTION:

A binary operation "*" on a set A is a function from A * A to A. i.e. *: A ×A →A. Hence *(a,b) = c, where a, b, c ∈A.

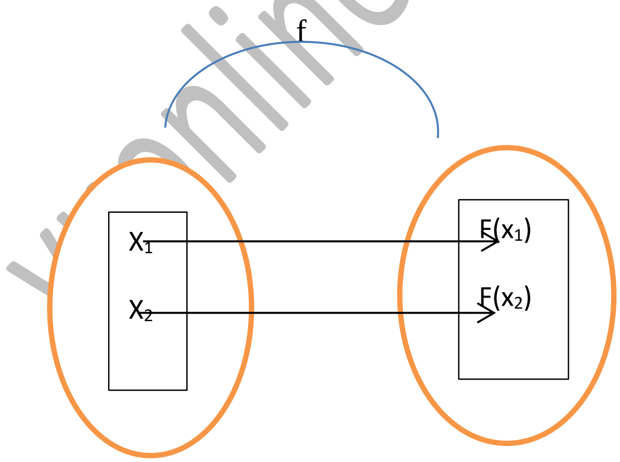
Lecture No.16 Types of functions

NJECTIVE

Let f: X →Y is a function. f is injective or one-to-one if, and only if, ∀ x1, x2 ∈X, if x1 ≠ x2 then f(x1) ≠ f(x2)That is.

ONE-TO-ONE FUNCTION

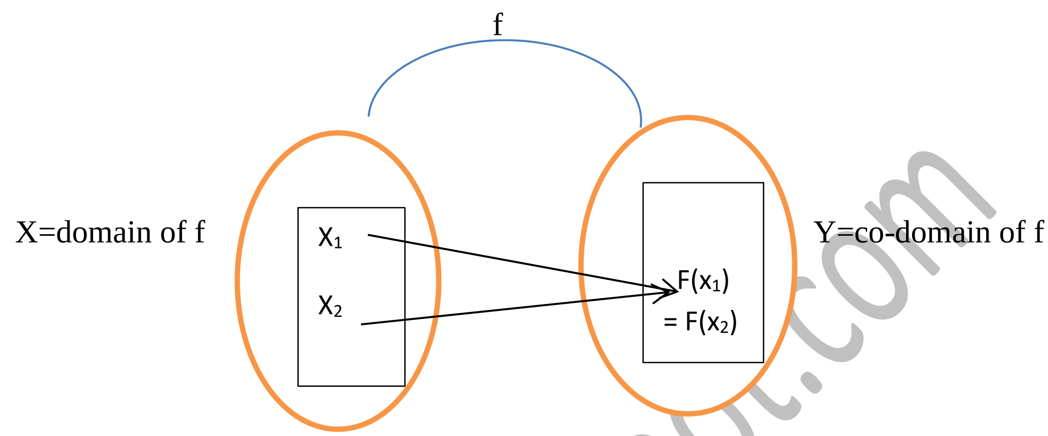
f is one-to-one if it maps distinct points of the domain into the distinct points of the co-domain.



A one-to-one function separates points.

FUNCTION NOT ONE-TO-ONE:

A function f: X →Y is not one-to-one iff there exist elements x1 and x2 in such that x1 ≠ x2 but f(x1) = f(x2).That is, if distinct elements x1 and x2 can found in domain of f that have the same function value.



A function that is not one-to-one collapses points together.

ALTERNATIVE DEFINITION FOR ONE-TO-ONE FUNCTION:

Denoted One-to-one by (1-1)

A function f: X →Y is one-to-one (1-1) iff ∀ x 1, x2 ∈X, if x1 ≠ x2 then f(x1) ≠ f(x2) (i.e. distinct elements of 1st set have their distinct images in 2nd set) The equivalent contra-positive statement for this implication is∀ x1, x2 ∈X, if f(x1) = f(x2), then x1 = x2

Example

Define f: R →R by the rule f(x) = 4x – 1

Let x1, x2 ∈R such that f(x1) = f(x2)

$\Rightarrow 4*X_1 - 1 = 4*X_2 - 1$

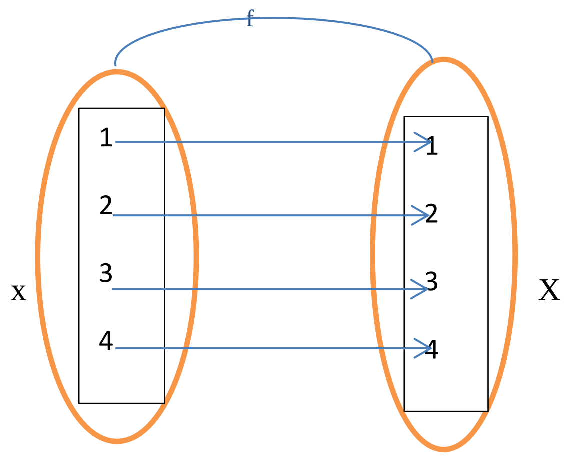
$\Rightarrow 4*x1 = 4*x2$

$\Rightarrow X_1 = X_2$

If $f(X_1) = f(X_2)$ then $X_1=X_2$ f is one-to-one

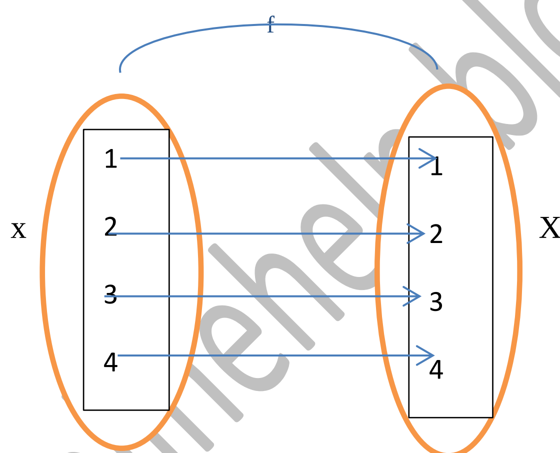
BIJECTIVE FUNCTION or ONE-TO-ONE CORRESPONDENCE

A function f: X→Y that is both one-to-one (injective) and onto (subjective) is called a bijective function or a one-to-one correspondence.



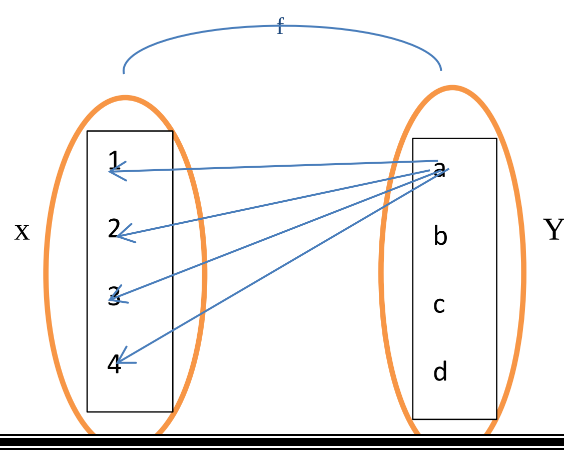
IDENTITY FUNCTION ON A SET:

Given a set X, define a function ix from X to X by ix(x) = x from all x ∈X. The function ix is called the identity function on X because it sends each element of X to itself.



CONSTANT FUNCTION:

A function f:X→Y is a constant function if it maps (sends) all elements of X to one element of Y i.e. ∀ x ∈X, f(x) = c, for some c ∈ Y



Lecture No.17 Inverse Function

EQUALITY OF FUNCTIONS:

Suppose f and g is functions from X to Y. Then f equals g, written f = g, if, and only if, f(x) = g(x) for all X∈X

EXAMPLE:

R→R by formulas:

Let

f(x) = |x| and

$g(x)=\sqrt{x^2}$

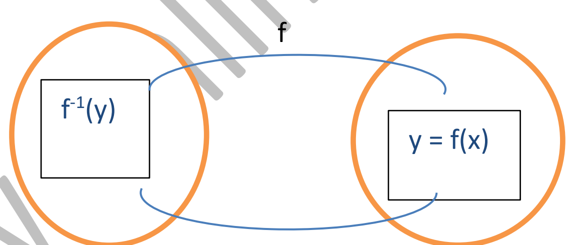
f(x) = |x|

INVERSE FUNCTION:

Suppose f: X→Y is a bijective function. Then the inverse function f-1: Y→X is defined

as: $\forall\ y \in Y, f^{-1}(y) = x \Leftrightarrow y = f(x)$

That is, f-1 sends each element of Y back to the element of X that it came from under f.



X=domain of f f^{-1} Y=co-domain of f

A function whose inverse function exists is called an invertible function.

COMPOSITION OF FUNCTIONS:

Let f: X →Y' and g: Y →Z be functions with the property that the range of f is a subset of the domain of g i.e. f(X) ⊆Y.

Define a new function gof:X →Z as follows:

(gof)(x) = g(f(x)) for all X ∈X The function gof is called the composition of f and g.

Lecture No.18 Composition of Functions

REAL-VALUED FUNCTIONS:

Let X be any set and R be the set of real numbers. A function f:X→R that assigns to each x∈X a real number f(x) ∈R is called a real-valued function.

If f: R →R, then f is called a real-valued function of a real variable.

SUM OF FUNCTIONS:

Let f and g be real valued functions with the same domain X. That is f:X →R and g:X →R. The sum of f and g denoted by f+g is a real valued function with the same domain X i.e. f+g: X →R defined by

(f+g)(x) = f(x) + g(x) ∀ x∈X

DIFFERENCE OF FUNCTIONS:

Let f: X →R and g: X →R be real valued functions. The difference of f and g denoted by f-g which is a function from X to R defined by (f-g)(x) = f(x) - g(x) ∀ x∈X

PRODUCT OF FUNCTIONS:

Let f: X →R and g:X →R be real valued functions. The product of f and g denoted f.g or simply fg is a function from X to R defined by

(f. g) (x) = f(x) . g(x) ∀ x∈X

QUOTIENT OF FUNCTIONS:

Let f: X→R and g: X →R be real valued functions. The quotient of f by g denoted by f/g is a function from X to R defined by

$$\frac{f}{g}(x) = \frac{f(x)}{g(x)}$$

SCALAR MULTIPLICATION:

Let f: X →R be a real valued function and c is a non-zero number. Then the scalar multiplication of f is a function c·f: R →R defined by (c·f)(x) = c·f(x) ∀x∈X

Let $f(x) = x2 + 1$ and $g(x) = x+2$ defines functions f and g from R to R.

Then

$(3f - 2g)(x) =(3f)(x) - (2g)(x)$

$=3 \cdot f(x) - 2 \cdot g(x)$

$= 3(x^2 +1) - 2 (x+2)$

$= 3x^2 - 2x-1$

$\forall x \in X$

FINITE SET:

A set is called finite if, and only if, it is the empty set or there is one-to-one correspondence from {1,2,3,…,n} to it, where n is a positive integer.

INFINITE SET:

A non empty set that cannot be put into one-to-one correspondence with {1,2,3,…,n}, for any positive integer n, is called infinite set.

CARDINALITY:

Let A and B be any sets. A has the same cardinality as B if, and only if, there is a one-toone correspondence from A to B.

COUNTABLE SET:

A set is count ably infinite if, and only if, it has the same cardinality as the set of positive integers Z+.

A set is called countable if, and only if, it is finite or count ably infinite.

A set that is not countable is called uncountable.

Lecture No.19 Sequence

SEQUENCE:

A sequence is just a list of elements usually written in a row.

EXAMPLES:

1. 1, 2, 3, 4, 5,…………..

2. 4, 8, 12, 16, 20, ……………

3. 2, 4, 8, 16, 32,……………….

4. 1, 1/2, 1/3, 1/4, 1/5, … ……….

5. 1, 4, 9, 16, 25, ………………….

6. 1, -1, 1, -1, 1, -1, ……..

The symbol "…" is called ellipsis, and reads "and so forth"

FORMAL DEFINITION:

A sequence is a function whose domain is the set of integers greater than or equal to a particular integer n_0.

Usually this set is the set of Natural numbers {1, 2, 3...} or the set of whole numbers {0, 1, 2, 3 …}.

ARITHMETIC SEQUENCE:

A sequence in which every term after the first is obtained from the preceding term by adding a constant number is called an arithmetic sequence or arithmetic progression (A.P.)

Common difference of A.P

The constant number, being the difference of any two consecutive terms is called the common difference of A.P., commonly denoted by "d".

GENERAL TERM OF AN ARITHMETIC SEQUENCE:

An arithmetic sequence is a sequence where the difference d between successive terms is constant. The general term of an arithmetic sequence can be written in terms of its first term a1, common difference d, and index n as follows: $an=a_1+(n-1)d$. An arithmetic series is the sum of the terms of an arithmetic sequence.

GEOMETRIC SEQUENCE:

A sequence in which every term after the first is obtained from the preceding term by multiplying it with a constant number is called a geometric sequence or geometric progression (G.P.)

Common ratio of the G.P

The constant number, being the ratio of any two consecutive terms is called the common ratio of the G.P. commonly denoted by "r".

GENERAL TERM OF A GEOMETRIC SEQUENCE:

Let a be the first tem and r be the common ratio of a geometric sequence. Then the sequence is a, ar^1, ar^2, ar^3 …

an = nth term = ar^{n-1} for all integers $n \geq 1$

SEQUENCES IN COMPUTER PROGRAMMING:

An important data type in computer programming consists of finite sequences known as one-dimensional arrays; a single variable in which a sequence of variables may be stored.
The names of k students in a class may be represented by an array of k elements “name” as: name [0], name [1], name [2]… name [k-1]

Lecture No.20 Series

SERIES:

The sum of the terms of a sequence forms a series. If a1, a2, a3, … represent a sequence of numbers, then the corresponding series is

$$a_1 + a_2 + a_3 + \cdots\cdots a_n = \sum_{k=1}^{\infty} a_k$$

SUMMATION NOTATION:

The capital Greek letter sigma $\sum$ is used to write a sum in a short hand notation. Where k varies from 1 to n represents the sum given in expanded form by = a1 + a2 + a3 + … + an

$$\sum_{k=1}^{n} a_k = a_m + a_{m+1} + a_{m+2} + \cdots\cdots a_n$$

Here k is called the index of the summation; m the lower limit of the summation and n the upper limit of the summation.

GEOMETRIC SERIES:

The sum of the terms of a geometric sequence forms a geometric series (G.S.).

For example 1 + 2 + 4 + 8 + 16 + …

Lecture No.21 Recursion I

Odd positive integer

An odd number is an integer when divided by two, either leaves a remainder or the result is a fraction. One is the first odd positive number but it does not leave a remainder 1.

1 is an odd positive integer

Some examples of odd numbers are 1, 3, 5, 7, 9, and 11. An integer that is not an odd number is an even number.

RECURSION:

Recursion is the process of defining a problem (or the solution to a problem) in terms of (a simpler version of) itself.

For example, we can define the operation "find your way home" as: If you are at home, stop moving.

RECURSIVELY DEFINED FUNCTIONS:

Recursively Defined Functions. A recursive definition of function f (·), defines a value of function at some natural number n in terms of the function's value at some previous point(s).

THE FACTORIAL OF A POSITIVE INTEGER:

Factorial, in mathematics, the product of all positive integers less than or equal to a given positive integer and denoted by that integer and an exclamation point.
Thus, factorial seven is written 7!, meaning 1 × 2 × 3 × 4 × 5 × 6 × 7. Factorial zero is defined as equal to 1.

$n! = n(n-1)!$

THE FACTORIAL FUNCTION DEFINED RECURSIVELY:

A recursive function is a no leaf function that calls itself. ... The factorial function can be rewritten recursively as factorial (n) = n × factorial (n – 1). The factorial of 1 is simply 1. Code Example 6.27

f(n-1). {(n! = n (n-1)!}

RECURRENCE RELATION:

A recurrence relation is an equation that defines a sequence based on a rule that gives the next term as a function of the previous term(s). ... For example, the recurrence relation $x_{n+1}=x_n+x_{n-1}$ can generate the Fibonacci numbers.

THE TOWER OF HANOI:

The puzzle was invented by a French Mathematician Adouard lucas in 1883.

The Tower of Hanoi is a mathematical game or puzzle. It consists of three rods and a number of disks of different diameters, which can slide onto any rod.

Example

Then m_n can be obtained recursively as follows

Let

$M_1 = 1$

$M_2 = 2$

$m_{k-1} + 1$ formula

$m_{k-1} + 1 = m_2 * m_1 + 1$

$m_2 * m_1 + 1 = 2*1+1$

$m_2 * m_1 + 1 = 3$

Lecture No.22 Recursion II

Parasitic Question